Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 4A nr. 20.5

$$
\begin{aligned}
&\int \cosh^3 x dx \\
&= \int \left(\frac{e^x + e^{-x}}{2}\right)^3 dx \\
&= \int \frac{\left(e^x + e^{-x}\right)^3}{8} dx \\
&= \frac{1}{8} \int \left(e^x + e^{-x}\right)^3 dx \\
&= \frac{1}{8} \int \left(e^{3x} + 3e^x + 3e^{-x} + e^{-3x}\right) dx \\
&= \frac{1}{8} \left(\int e^{3x} dx + \int 3e^x dx + \int 3e^{-x} dx + \int e^{-3x} dx\right) \\
&= \frac{1}{8} \left(\frac{1}{3} e^{3x} + 3e^x - 3e^{-x} - \frac{1}{3} e^{-3x}\right) + C \\
&= \frac{1}{8} \left(\frac{1}{3}\left(e^{3x} - e^{-3x}\right) + 3\left(e^x - e^{-x}\right)\right) + C \\
&= \frac{1}{8} \left(\frac{1}{3} \cdot 2 \sinh 3x + 3 \cdot 2 \sinh x\right) + C \\
&= \frac{1}{12} \sinh 3x + \frac{3}{4} \sinh x + C
\end{aligned}
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References